

NN



Neural Network

⇒ Neural Network:

→ What is a Neuron?

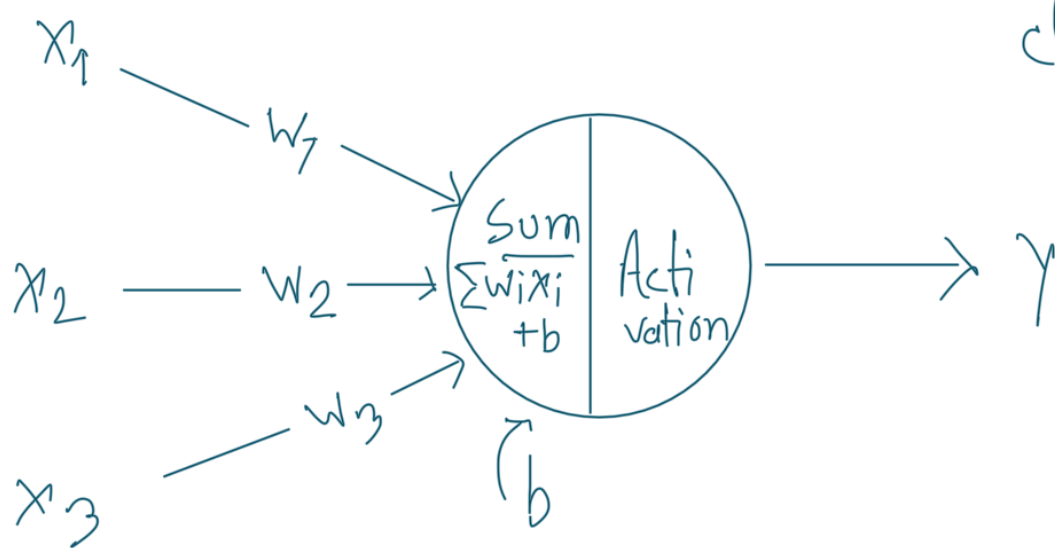
You can consider a neuron as a node which takes one/more input and provides output based on some calculation.

→ Neural Network?

If multiple neurons (nodes) are connected with each other then we can consider it as a network of neurons; aka neural networks.

→ What is a perceptron?

Perceptron is a one layer neural network that can classify things into two parts.  
[Linear binary classifier]

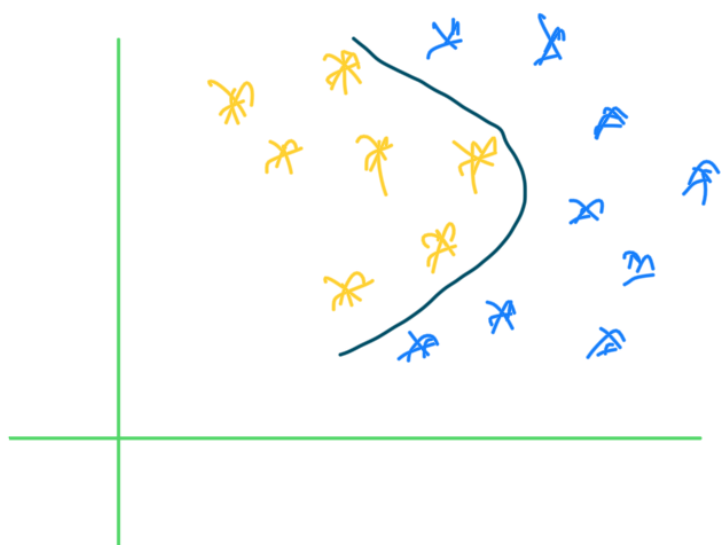


Model → Output

Input  $\rightarrow$  Process

Inputs are multiplied by weights (indicates the importance)  $\rightarrow$  Sum them up along with a static number  $b$  (bias)  $\rightarrow$  Pass it to an activation function to map the output (0/1).

$\rightarrow$  Multi Layer Perception (MLP)?



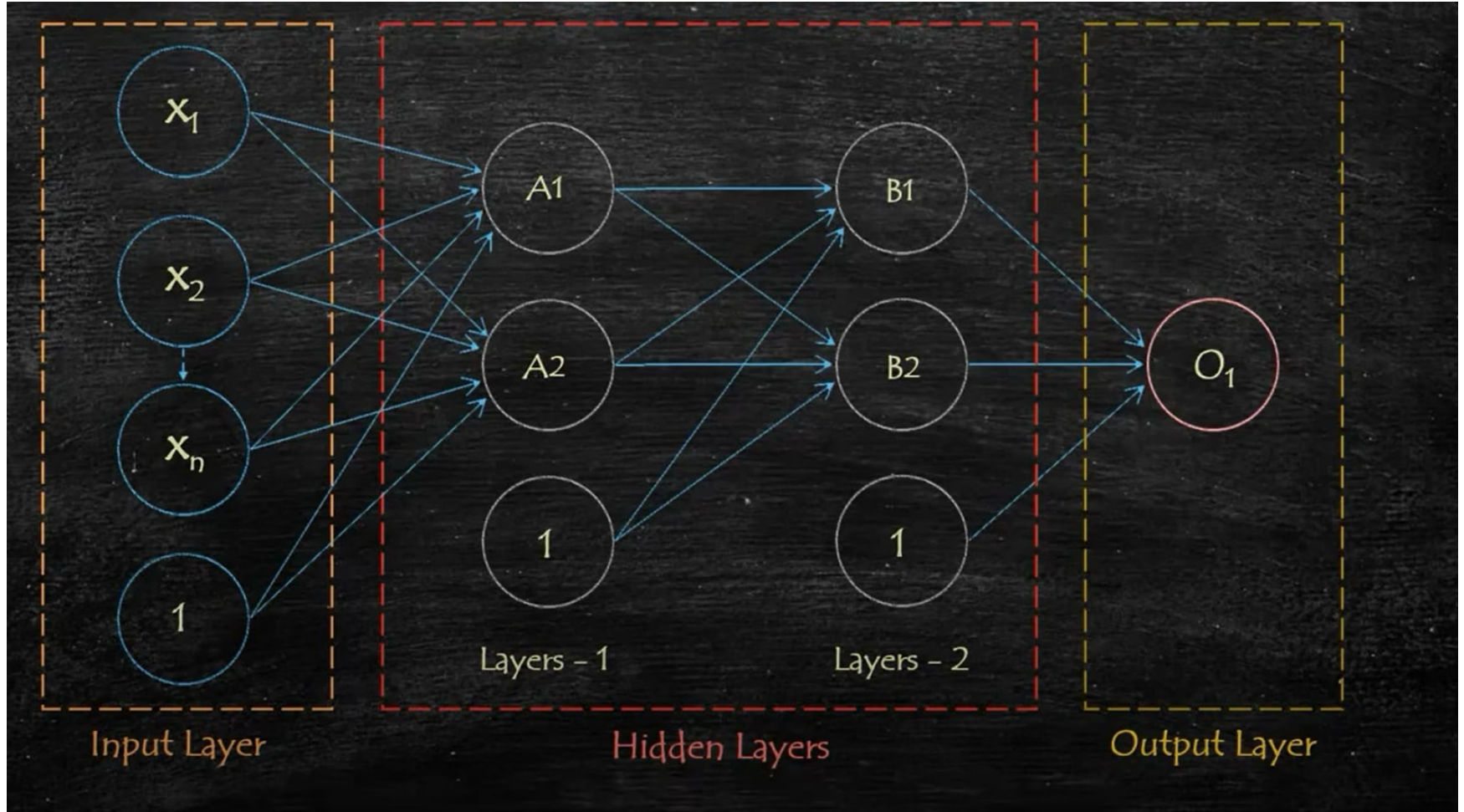
$\rightarrow$  A non-linear separable problem.

$\rightarrow$  This is where MLP helps us.

A network of layer based perceptions. that means connections of perceptions among multiple layers.

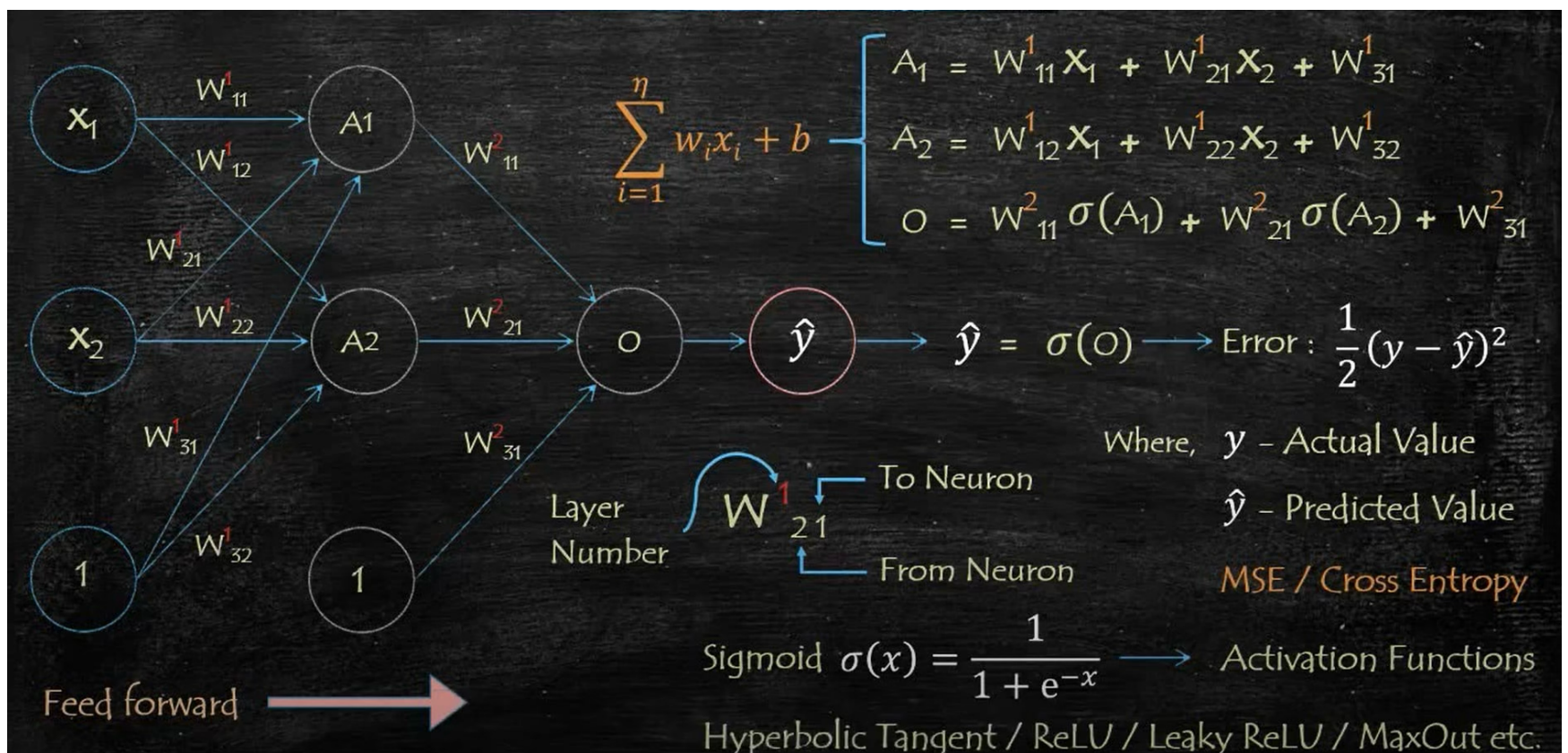
$\rightarrow$  A fully connected network of MLP





→ In the picture above ① is bias.

## Forward / Feed-Forward Propagation





⇒ In this example  $w'_{31}/w'_{32}/w^2_{31}$  are the bias value.

### Mathematical representation of feed-forward Network

Transposed Weight Vector & Matrix

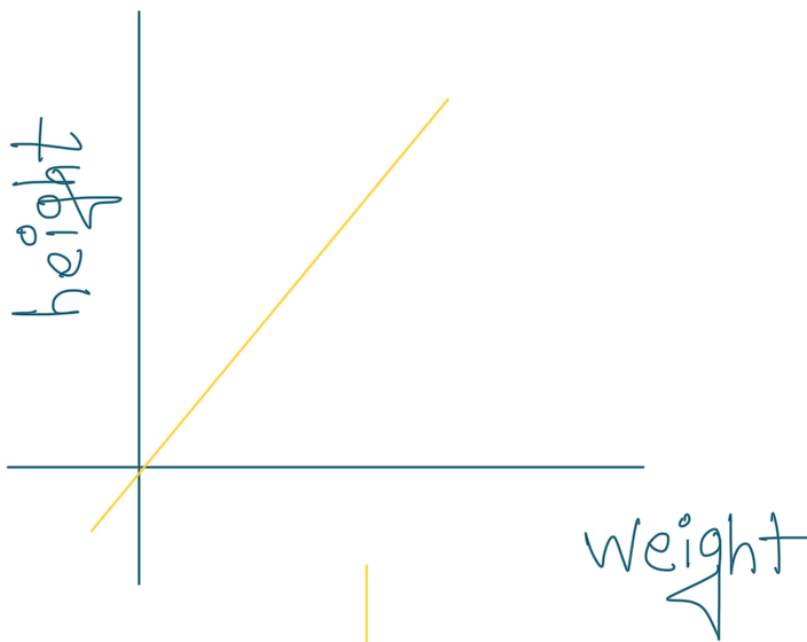
$$\hat{y} = \sigma \cdot \begin{bmatrix} w^2_{11} & w^2_{21} & w^2_{31} \end{bmatrix} \cdot \sigma \cdot \begin{bmatrix} w^1_{11} & w^1_{21} & w^1_{31} \\ w^1_{12} & w^1_{22} & w^1_{32} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$$

Column Vector of Input values

$$\hat{y} = \sigma \cdot w^2 \cdot \sigma \cdot w^1 X$$

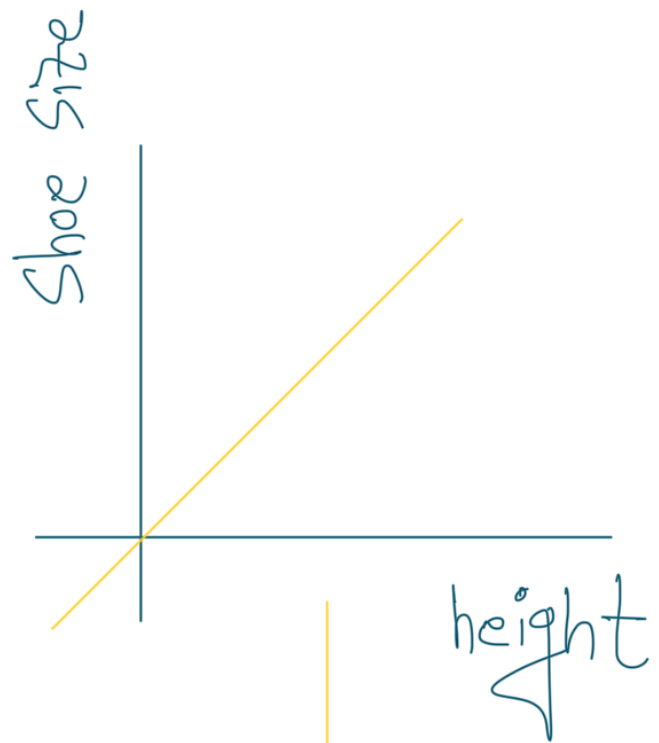


Chain Rule:



Change of height  
with respect to weight

$$\frac{d \text{ height}}{d \text{ weight}}$$



Change of shoe size  
with respect to height

$$\frac{d \text{ shoe size}}{d \text{ height}}$$

⇒ Change of shoe size with respect to  
weight.

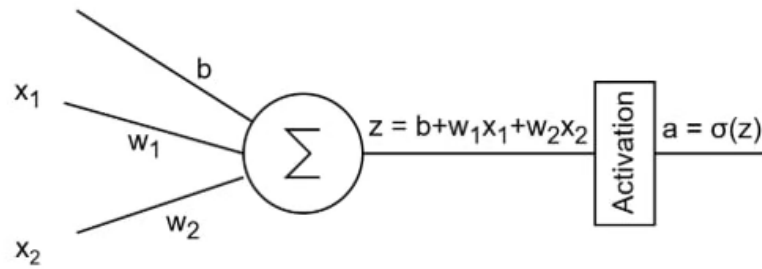
$$\frac{d \text{ Shoe Size}}{d \text{ Weight}} = \frac{d \text{ Shoe size}}{d \text{ Height}} * \frac{d \text{ height}}{d \text{ weight}}$$

Shoe Size connects Height and  
Height connects Weights.

And this is chain rule.

→ learn more about

→ Now let's look it for neural networks:



$$L = -y \log a - (1 - y) \log(1 - a)$$

$$\text{Where, } a = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\text{Where, } z = x_1 w_1 + x_2 w_2 + b$$

To update the weight  $w_1$ , we need to perform  $w_1 = w_1 - \frac{dL}{dw_1}$

All these individual functions are nested to each other.

Thus, to calculate the derivative  $\frac{dL}{dw_1}$ , according to the chain rule,

First, we need to calculate  $\frac{dL}{da}$  [Derivative of L with respect to a]

Then, we need to calculate  $\frac{da}{dz}$  [Derivative of a with respect to z]

Afterward, we need to calculate  $\frac{dz}{dw_1}$  [Derivative of  $W_1$  with respect to w]

Finally,  $\frac{dL}{dw_1}$  will be, according to the chain rule:

$$\frac{dL}{dw_1} = \frac{dz}{dw_1} * \frac{da}{dz} * \frac{dL}{da}$$



→ For a larger network:

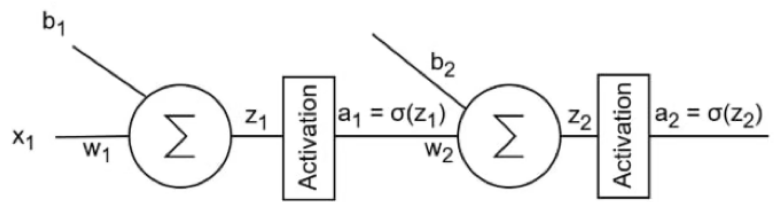
$$L = -y \log a_2 - (1 - y) \log(1 - a_2)$$

$$\text{Where, } a_2 = \sigma(z_2) = \frac{1}{1 + e^{-z_2}}$$

$$\text{Where, } z_2 = a_1 w_2 + b_2$$

$$\text{Where, } a_1 = \sigma(z_1) = \frac{1}{1 + e^{-z_1}}$$

$$\text{Where, } z_1 = x_1 w_1 + b_1$$



At first, to update the weight  $w_2$ , we need to perform  $w_2 = w_2 - \frac{dL}{dw_2}$

All these individual functions are nested to each other.

Thus, to calculate the derivative  $\frac{dL}{dw_2}$ , according to the chain rule,

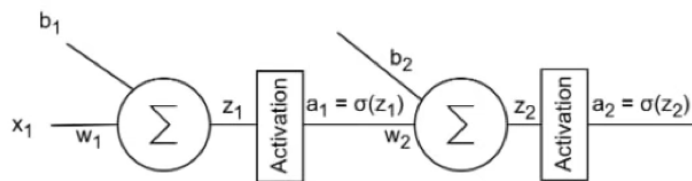
First, we need to calculate  $\frac{dL}{da_2}$  [Derivative of L with respect to  $a_2$ ]

Then, we need to calculate  $\frac{da_2}{dz_2}$  [Derivative of  $a_2$  with respect to  $z_2$ ]

Afterward, we need to calculate  $\frac{dz_2}{dw_2}$  [Derivative of  $z_2$  with respect to  $w_2$ ]

Finally,  $\frac{dL}{dw_2}$  will be, according to the chain rule:

$$\frac{dL}{dw_2} = \frac{dz_2}{dw_2} * \frac{da_2}{dz_2} * \frac{dL}{da_2}$$



Meanwhile, to update the weight  $w_1$ , we need to perform  $w_1 = w_1 - \frac{dL}{dw_1}$

To calculate the derivative  $\frac{dL}{dw_1}$ , according to the chain rule,

First, we need to calculate  $\frac{dL}{da_2}$  [Derivative of L with respect to  $a_2$ ]

Then, we need to calculate  $\frac{da_2}{dz_2}$  [Derivative of  $a_2$  with respect to  $z_2$ ]

Afterward, we need to calculate  $\frac{dz_2}{da_1}$  [Derivative of  $z_2$  with respect to  $a_1$ ]

In the next step, calculate  $\frac{da_1}{dz_1}$  [Derivative of  $a_1$  with respect to  $z_1$ ]

Next, calculate  $\frac{dz_1}{dw_1}$  [Derivative of  $z_1$  with respect to  $w_1$ ]

Finally,  $\frac{dL}{dw_1}$  will be, according to the chain rule:

$$\frac{dL}{dw_1} = \frac{dz_1}{dw_1} * \frac{da_1}{dz_1} * \frac{dz_2}{da_1} * \frac{da_2}{dz_2} * \frac{dL}{da_2}$$

→ Why don't we differentiate  $z_2$  with respect to  $w_2$ ?

□ Because we are finding the change of L with respect to  $w_1$ .

And  $w_1$   $w_2$  are independent to each other.

change in  $w_1$  does not effect  $w_2$  but does effect  $a_1 \rightarrow$  which flows into  $z_2$ .

Now let's say we have three nested weights (layers)  $w_1, w_2, w_3$  and we want to find  $\frac{dL}{dw_1}$ .

$$\Rightarrow \frac{dL}{dw_1} = \frac{dL}{da_3} \times \frac{da_3}{dz_3} \times \frac{dz_3}{da_2} \times \frac{da_2}{dz_2} \times \frac{dz_2}{da_1} \times \frac{da_1}{dz_1} \times \frac{dz_1}{dw_1}$$

## Backpropagation

Backpropagation = Chain Rule + Updating weights and bias in each layer.

